

$$G_{\alpha\beta} = 8\pi G T_{\alpha\beta}$$

$$H^2 = \frac{1}{3M_{Pl}^2} \rho + \frac{\Lambda}{3}$$

$$L \propto V_{rot}^{\alpha}$$

$$Z_{SFR} \propto (Z_{gas})^{\eta}$$

Symbolic Regression in the Physical Sciences

Zobrist Hash-Based Duplicate Detection in Symbolic Regression

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$$\vec{F} = m\vec{a}$$

$$K = Ae^{-Kl}$$

$$\eta = k m \alpha$$

$$\frac{\partial \mathcal{L}}{\partial t} = D \frac{\partial^2 \mathcal{L}}{\partial z^2} + s s \frac{\partial \mathcal{L}}{\partial z}$$

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$$

About Me

Professor of Data Science and Machine Learning

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Research Interests

- symbolic regression, physical modeling, simulation and optimization
- explainability and interpretable machine learning
- algorithm design, scalability, energy-efficient AI
- author of the Operon library

<https://github.com/heal-research/operon>

Zobrist Hashing

Technique used in abstract board games to implement transposition tables.

- Starts by randomly generating bitstrings for each possible element of a board game (i.e. for each combination of a piece and a position)
- The Zobrist hash is computed by combining those bitstrings using bitwise XOR
- If the bitstrings are long enough, the probability of collisions is negligible

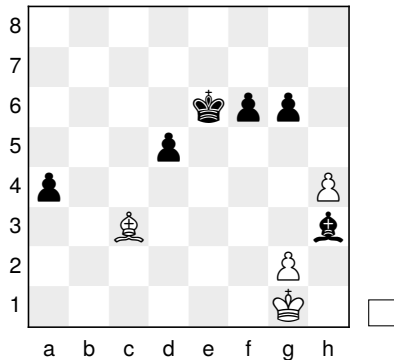
Bitwise XOR → incremental hash update

$$a \oplus b = b \oplus a \quad (\text{associativity})$$

$$a \oplus (b \oplus c) = (a \oplus b) \oplus c \quad (\text{commutativity})$$

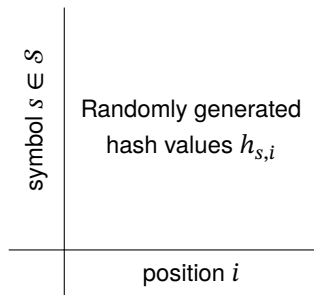
$$a \oplus b \oplus b = a \quad (\text{involution})$$

Zobrist Hashing



	2173	1288	1235	2880	3217
	1074	1570	3800	1538	1260
	3658	1022	1433	1801	2504
	4054	3302	2029	1632	2549
⋮	⋮	⋮	⋮	⋮	⋮

Zobrist Hashing



$$\mathcal{H}(\text{state}) = \bigoplus h_{s,i}$$

If the **state** of a program¹ can be described by a list of symbols and their associated positions, then it can be efficiently maintained using Zobrist hashing.

¹algorithm, search method, simulation, optimization procedure

Fast recall of state-history in kinetic Monte Carlo simulations utilizing the Zobrist key

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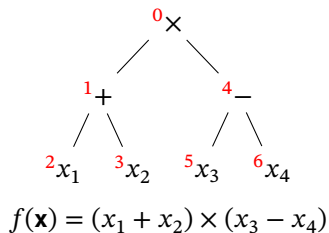
Available online 11 November 2004

Abstract

We present a novel application of the Zobrist hashing method, known in the computer chess literature, to simulation of diffusional phase transformations in metal alloys. A history of previously visited states can be easily maintained, allowing very fast lookup of energies and transition rates calculated earlier in the simulation. The method has been applied to the simulation

Zobrist Hashing

Expression tree T



Prefix representation:

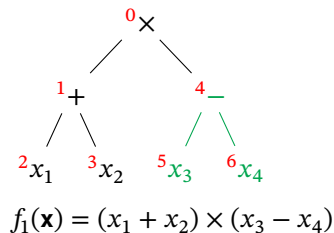
$\times + x_1 x_2 - x_3 x_4$
 0 1 2 3 4 5 6

$$\mathcal{H}(T) = h_{\times,0} \oplus h_{+,1} \oplus h_{x_1,2} \oplus h_{x_2,3} \oplus h_{-,4} \oplus h_{x_3,5} \oplus h_{x_4,6}$$

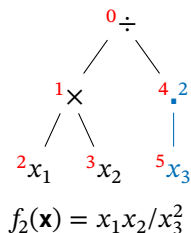
+	Filled with random values
−	
×	
÷	
⋮	
x_1	
x_2	
x_3	
x_4	
c	
position $i = 0, \dots, N$	

Zobrist Hashing in Symbolic Regression

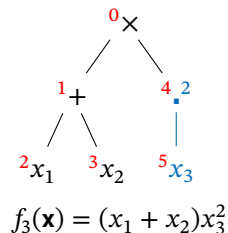
Parent tree T_1



Parent tree T_2



Child tree C_1



Incremental hash update after crossover

$$\mathcal{H}(C_1) = \mathcal{H}(T_1) \oplus \underbrace{h_{-,4} \oplus h_{x_3,5} \oplus h_{x_4,6}}_{\text{XOR "out" } (x_3 - x_4)} \oplus \underbrace{h_{.,4} \oplus h_{x_3,5}}_{\text{XOR "in" } x_3^2}$$